CS1910 Computer Science I • Dictionaries

Partner workbook

Ciphers, Pictures and Food Chains

In lecture with a partner; challenge at home • about 40 minutes

Name: _____ **Partner:** _____

This PDF can be filled in: click in an answer space to type, or print it and write. Draw diagrams with your PDF viewer's pen or markup tool, or on paper.

What this workbook practises

A dictionary maps **keys** to **values**, so you can look things up directly instead of searching. In each problem, decide first what the keys and the values should be.

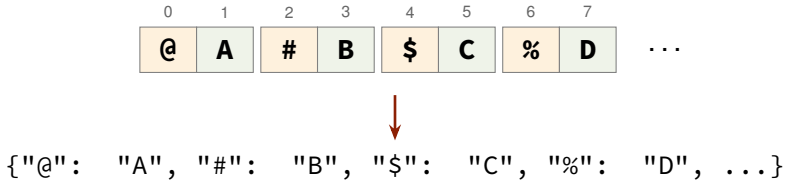
1 Ciphers

about 12 minutes

The problem

You are given a list that represents a cipher. **Even** positions hold symbols and **odd** positions hold their meanings (one symbol for each capital letter):

```
["@", "A", "#", "B", "$", "C", "%", "D", "&", "E", ..., "'", "Y", "\"",
                                                              "Z"]
```



1. What should the keys of the decoding dictionary be, and what should the values be? Why is a dictionary better than the list for *decoding*?

2. Write `transform_cipher(cipher_list)`, which returns the dictionary.

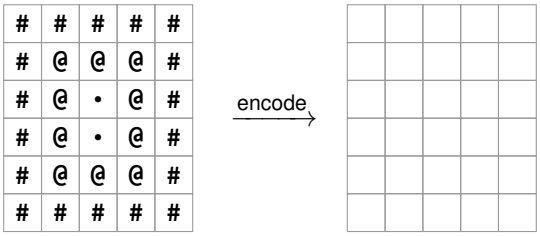
3. **Generalization:** we do not want anyone to know how the cipher is stored. Write `encode(message)`, which uses the *list* inside, and `decode(symbols)`, which uses a *dictionary* inside.

2 Storing a picture

about 10 minutes

The problem

A 2D picture has a boundary (an outline of a character, a coastline on a map, . . .). Three characters describe it: `@` is the boundary, `#` is outside, and `.` is inside. To send it, we encode `@` as 1, `#` as 0 and `.` as 2.



4. Fill in the encoded grid. How would you store the picture? How would you store the mapping between characters?

5. Sketch the algorithm to create the encoding, then translate it into Python.

3 Challenge (at home): the food chain

about 15 minutes

The problem

Each line lists a predator first, then the prey it eats. *Producers* eat no other creature; *apex predators* are eaten by no one; some creatures eat only producers (*vegetarians*).

```
Bird,Prawn,Mussels,Crab,Limpets,Whelk
Crab,Mussels,Limpets
Fish,Prawn
Limpets,Seaweed
Zooplankton,Phytoplankton
Lobster,Crab,Mussels,Limpets,Whelk
Mussels,Phytoplankton,Zooplankton
Prawn,Zooplankton
Whelk,Limpets,Mussels
```

6. How would you transform these lines into a dictionary? What are the keys and values? Do you need anything else?

7. Use your dictionary to answer each question.

Question	Answer
the apex predators	
the producers	
the tastiest (most eaten)	
the omnivores (eat producers and non-producers)	

✓ Before you leave

about 1 minutes

- ☐ I can choose sensible keys and values for a dictionary.
- ☐ I can build a dictionary with a loop, and use it to look things up.
- ☐ I can hide how data is stored behind functions.